VIKRAM S

B.Tech - Computer Science and Engineering

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PROFESSIONAL SUMMARY

7th-semester Computer Science undergraduate (CGPA 9.53), available for a full 8th-semester internship, who builds GenAI-powered software end to end: LLM agents, RAG and semantic search, LLM fine-tuning, and production backend and full-stack systems. As an SDE Intern at BNY, built an LLM agent with MCP that automated exception triage previously handled by 2 engineers on a platform serving 100+ developers. Works AI-first across the development lifecycle, using Claude Code and agentic tooling from design through code generation and testing. Strong in OOP with Python, Java and TypeScript, SQL, and cloud deployment.

EDUCATION

SRM Institute of Science and Technology **Aug 2023 - Jul 2027**

B.Tech, Computer Science and Engineering | CGPA: 9.53 / 10 | Currently in 7th semester Chennai

WORK EXPERIENCE

SDE Intern | BNY **Jun 2026 - Aug 2026**

Software Engineering, Agile team | On-site, Chennai

* Eliminated a manual exception-triage workflow staffed by 2 engineers on a platform serving 100+ developers, by building an LLM-powered agent in Java and Spring Boot that traces each Splunk error to the faulty GitLab code and, via Jira, to the responsible developer and team.
* Unified 3 enterprise systems (Splunk, GitLab, Jira) into one automated pipeline, by integrating them as MCP servers and applying LLM reasoning at each stage, from error trace to source code to code owner.
* Aligned the team on the agent architecture before build, by authoring its technical design document and developing it with an AI-assisted workflow using Claude Code.
* Modernised the end-to-end test suite, by migrating tests from Selenium to Playwright and delivering clean, testable code that followed secure coding practices.

Tech: Java, Spring Boot, LLMs, Claude Code, MCP, REST APIs, Splunk, GitLab, Jira, Playwright, Selenium

AI/ML Research Intern | King Faisal University **Jul 2025 - Jun 2026**

Saudi Arabia | Remote

* Achieved 94.4% accuracy and AUROC > 0.99 on 9-class waste classification (TrashNeXt), by designing a hybrid framework that stacks ViT/CNN backbones (Swin-Base + EfficientNetV2-S) under an XGBoost meta-learner; a second variant (LeViT-256 + EfficientNetV2-S) reached 90.3%.
* Selected the best configuration across 50+ trials with 5-run validation, by running Optuna TPE search, and explained misclassifications by applying SHAP and LIME for interpretability and error analysis.
* Co-authored a peer-reviewed journal publication, after optimising the pipeline for low-latency, edge-deployable inference.

Tech: PyTorch, Vision Transformers, CNNs, XGBoost, Optuna, SHAP, LIME, ClearML

PROJECTS

OutBox - Distributed Email Job Scheduler **GitHub | Live**

* Scheduled 1,000 emails in ~0.5 s and delivered all 1,000 in a load test with zero duplicates and one attempt each, by modelling every email as a delayed BullMQ job, with bulk Postgres inserts and bulk queueing.
* Enforced per-sender and per-campaign hourly limits across any number of workers without dropping a job, by writing an atomic Redis Lua slot-reservation rate limiter that books overflow into the next window; verified by unit tests including 50 concurrent reservations against real Redis.
* Guaranteed no email is sent twice through retries, restarts and racing workers, by layering idempotency guards: deterministic job IDs, per-email Redis locks, conditional Postgres status claims, and a delivered marker.
* Delivered typo-tolerant, search-as-you-type email search, by indexing in Elasticsearch with a custom address analyzer, fuzzy and prefix queries, and a graceful Postgres fallback; shipped Google OAuth 2.0 login, Slack OAuth alerts and a React 19 dashboard, deployed on Render and Vercel.

Tech: TypeScript, Express, BullMQ, Redis, PostgreSQL (Drizzle ORM), Elasticsearch, React, TanStack Query, Tailwind, Docker, Vitest

Local RAG Pipeline - Document Q&A with Semantic Search **GitHub**

* Answered questions over private documents fully on-device with source-cited responses, by building a modular ingestion -> embedding -> retrieval -> generation pipeline with Llama 3.2 via Ollama (streaming and non-streaming).
* Ingested 6 document formats (PDF, DOCX, HTML, CSV, TXT, MD) into predictable token budgets, by engineering a token-aware recursive chunker (tiktoken, 512 tokens / 64 overlap) with SHA-256 fingerprinting for idempotent re-ingestion.
* Supplied relevant, non-redundant context to the LLM, by implementing dense retrieval over a persistent ChromaDB cosine index with MMR re-ranking (20 candidates -> top 5) and metadata filtering.

Tech: Python, sentence-transformers (all-MiniLM-L6-v2), ChromaDB, Ollama, tiktoken, Pydantic + YAML config, Click + Rich CLI

SprintBoard - RL Environment for LLM Agents (OpenEnv Hackathon 2026) **GitHub | HF Space**

* Raised the mean grader score on 15 sprint-planning tasks from 0.39 (random-action baseline) to 0.99, by fine-tuning Qwen2.5-1.5B-Instruct with LoRA through an SFT warm-start followed by TRL GRPO against the live environment.
* Created a training arena where an LLM acts as Scrum Master with 18 free-form commands and a 15-step budget, by building an OpenEnv-compliant environment scored by a deterministic 3-axis grader (investigation, planning quality, process).
* Made the work reproducible, by shipping a Dockerised Hugging Face Space, a Colab training notebook and a published LoRA adapter.

Tech: Python, PyTorch, Hugging Face TRL (GRPO), LoRA, FastAPI, Docker, OpenEnv

NEXUS - SRM Academia Portal **GitHub | Live**

* Served 10K+ visitors with one platform unifying attendance, marks, timetable, CGPA and courses, by building and deploying a React + Vite progressive web app on Vercel with modern UI/UX.
* Turned the university portal unstructured pages into clean JSON APIs, by engineering Express REST endpoints with Cheerio scrapers.
* Shipped 5+ backend services (Supabase persistence, authentication, push notifications, cron jobs, PWA support) instrumented with Vercel Analy

Tech: React, Vite, Node.js, Express, Supabase, Cheerio, Vercel

Airline Route Optimization - Weather-Aware Routing **GitHub**

* Recommended direct, one-stop and two-stop routes ranked by distance and live weather risk, by modelling routes as a weighted NetworkX graph and re-scoring paths with OpenWeatherMap data.
* Exposed route search and analytics to an interactive map dashboard, by building a Flask REST API with PySpark CSV processing & Leaflet + EC

Tech: Python, Flask, PySpark, NetworkX, Pandas, JavaScript, Leaflet.js, ECharts

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL

GenAI & AI-Assisted SDLC: LLM agents, MCP, RAG, Claude Code, LoRA fine-tuning, GRPO (TRL), HF Transformers, Ollama, ChromaDB

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Optuna, SHAP / LIME, Computer Vision

Backend & Full-Stack: Spring Boot, Node.js, Express, FastAPI, Flask, React, Angular, Vite, Tailwind CSS, HTML/CSS, REST APIs

Data & Search: PostgreSQL, MySQL, Supabase, Redis, Elasticsearch, BullMQ, PySpark

Cloud, DevOps & Testing: Docker, CI/CD (Jenkins), Git, GitHub, GitLab, Linux, Vercel, Render, HF Spaces, Playwright, Selenium, Vitest

CS Fundamentals: Object-Oriented Programming & Design, Data Structures & Algorithms, Operating Systems, DBMS

AWARDS & ACHIEVEMENTS

1st Place - IEEE E-JIGYASA Project Expo 2025 **Apr 2025**

* Won 1st place at the national-level IEEE Student Chapter Innovation Expo, by representing SRM IST with a smart home automation system powered by cloud-integrated IoT and machine learning.

IIT Madras - Walmart Center for Tech Excellence Workshop **Apr 2025**

* Achieved high recall on critical-failure detection, by designing a sensor-based failure-prediction pipeline (ingestion -> preprocessing -> feature engineering -> real-time LSTM inference) with a dashboard alerting system.